24. Some meta-directing substituents in aromatic substitution are given. Which one is most deactivating ?

a) – COOH b) – NO_2
c) – C ≡ N d) – SO_3H

25. Which of the following can be used as the halide component for friedal - crafts reaction ?

a) Chloro benzene

b) Bromo benzene

c) chloro ethene

d) isopropyl chloride

26. An alkane is obtained by decarboxylation of sodium propionate. Same alkane can be prepared by

a) Catalytic hydrogenation of propene

b) action of sodium metal on iodomethane

c) reduction of 1 – chloro propane

d) reduction of bromomethane

27. Which of the following is aliphatic saturated hydrocarbon

a) C_8H_{18} b) C_9H_{18}
c) C_8H_{14} d) All of these

28. Identify the compound 'Z' in the following reaction

$$C_2H_6O \xrightarrow[623\ K]{Al_2O_3} X \xrightarrow{O_3} Y \xrightarrow{Zn/H_2O} (Z)$$

a) Formaldehyde

b) Acetaldehyde

c) Formic acid d) none of these

29. Peroxide effect (Kharasch effect) can be studied in case of

a) Oct – 4 – ene b) hex – 3 – ene
c) pent – 1 – ene d) but – 2 – ene

30. 2 – butyne on chlorination gives

a) 1 – chloro butane
b) 1, 2 – dichloro butane

c) 1, 1, 2, 2 – tetrachlorobutane
d) 2, 2, 3, 3 – tetra chloro butane

II. Write brief answer to the following questions.

31. Give IUPAC names for the following compounds

1) $CH_3–CH=CH–CH=CH–C\equiv C–CH_3$

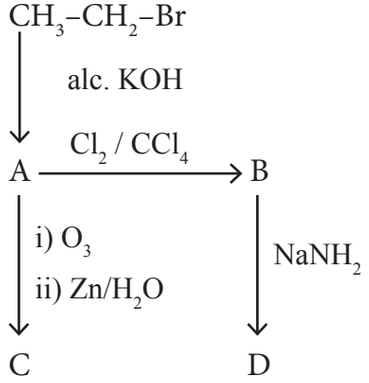
2) $CH_3–\underset{\displaystyle CH_3}{\overset{\displaystyle C_2H_5}{\underset{|}{\overset{|}{C}}}}–\underset{\displaystyle H}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{C}}}}–C\equiv C–CH_3$

3) $(CH_3)_3C–C\equiv C–CH(CH_3)_2$

4) ethyl isopropyl acetylene

5) $CH\equiv C–C\equiv C–C\equiv CH$

32. Identify the compound A, B, C and D in the following series of reactions



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